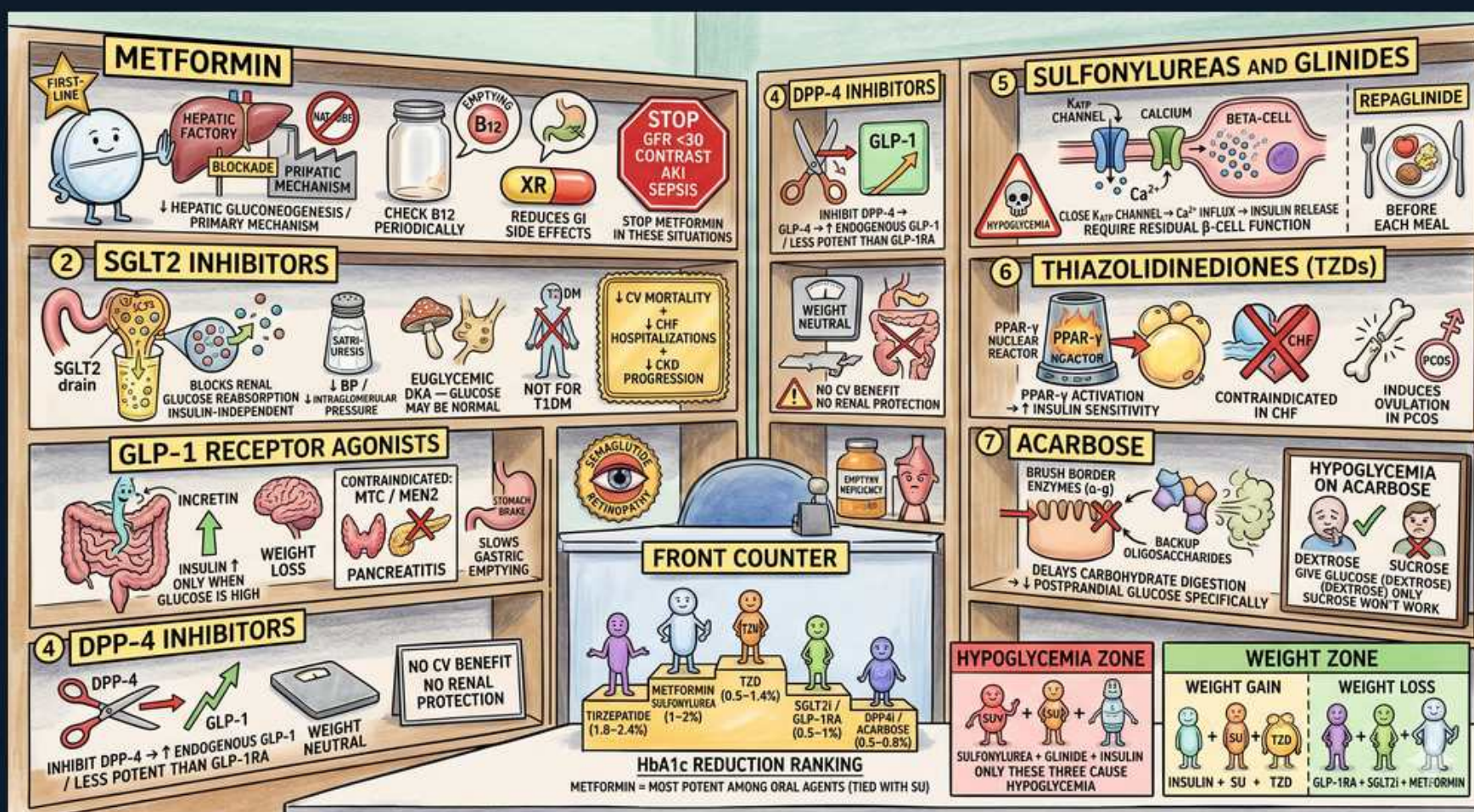


Oral & Injectable Antidiabetics — Mechanism, Renal & CV Effects



1. Metformin — first line

WHAT YOU SEE

Panel 1, liver + biguanide pills, FIRST LINE banner

WHAT IT ENCODES

Decreases hepatic gluconeogenesis (primary mechanism), no weight gain, no hypoglycemia. Largest HbA1c drop among oral agents (~1.5%). First-line for type 2 DM unless contraindicated.

BOARD TRAP

Does NOT cause hypoglycemia as monotherapy — that points to sulfonylurea or insulin instead.

2. Metformin — B12 & GI

WHAT YOU SEE

B12 icon + GI/reduced absorption symbols

WHAT IT ENCODES

Causes chronic B12 deficiency — check B12 periodically. GI side effects (diarrhea, nausea) are most common and dose-limiting. Take with meals.

BOARD TRAP

New peripheral neuropathy in a metformin patient may be B12 deficiency, not just diabetic neuropathy.

3. Metformin — lactic acidosis stops

WHAT YOU SEE

STOP sign: GFR<30, contrast, AKI, sepsis

WHAT IT ENCODES

Hold for eGFR <30, iodinated contrast, AKI, sepsis, hypoxia — risk of lactic acidosis. Stop before situations that impair renal clearance or oxygenation.

BOARD TRAP

Contrast + continued metformin with low GFR is the classic lactic acidosis vignette.

4. SGLT2 inhibitors — mechanism

WHAT YOU SEE

Panel 2, proximal tubule 'SGLT2 drain', glucose in urine

WHAT IT ENCODES

Block proximal tubule glucose reabsorption causing glucosuria. Lower BP and weight modestly. Insulin-independent action.

BOARD TRAP

Glucosuria with normal/low serum glucose is expected on SGLT2i — not a UTI or renal glucosuria disorder by itself.

5. SGLT2i — CV/renal benefit

WHAT YOU SEE

Heart + kidney, lower CV death/HF/CKD progression

WHAT IT ENCODES

Reduce CV death, heart failure hospitalizations, and CKD progression — give in HFrEF and diabetic kidney disease. Not for type 1 DM.

BOARD TRAP

Euglycemic DKA is the SGLT2i trap — DKA with near-normal glucose; do not exclude DKA on glucose alone.

6. GLP-1 receptor agonists

WHAT YOU SEE

Panel 3, gut/incretin, weight loss icon

WHAT IT ENCODES

Incretin mimetics: increase glucose-dependent insulin, decrease glucagon, slow gastric emptying, promote satiety/weight loss. Proven CV benefit. Low hypoglycemia risk alone.

BOARD TRAP

Contraindicated with personal/family history of medullary thyroid cancer or MEN2 — that is the discriminator vs other agents.

7. GLP-1 — pancreatitis/gastric

WHAT YOU SEE

Pancreas + stomach warning icons

WHAT IT ENCODES

Adverse effects: nausea, gastroparesis-like delayed emptying, and pancreatitis signal. Use caution with prior pancreatitis.

BOARD TRAP

Persistent vomiting + epigastric pain on a GLP-1 agonist suggests pancreatitis, not simple GI intolerance.

8. DPP-4 inhibitors

WHAT YOU SEE

Panel 4, '-gliptin', weight neutral, no CV benefit

WHAT IT ENCODES

Block DPP-4 increasing endogenous GLP-1 — weight neutral, low hypoglycemia, modest A1c effect. No proven CV benefit, no renal protection.

BOARD TRAP

Weak agent chosen when a vignette expects weight loss or CV benefit — those favor GLP-1/SGLT2, not gliptins.

9. Sulfonylureas / glinides

WHAT YOU SEE

Panel 5, beta-cell K⁺ channel close, repaglinide

WHAT IT ENCODES

Close beta-cell K-ATP channels causing depolarization, Ca influx, and insulin release — glucose-independent. Glinides act before each meal, shorter acting.

BOARD TRAP

Sulfonylurea is the classic cause of prolonged hypoglycemia, especially in renal impairment/elderly — admit and observe.

10. Thiazolidinediones (TZDs)

WHAT YOU SEE

Panel 6, PPAR-gamma nucleus, insulin sensitivity

WHAT IT ENCODES

PPAR-gamma agonists increase peripheral insulin sensitivity. Useful in NAFLD/PCOS. Slow onset.

BOARD TRAP

Contraindicated in CHF — cause fluid retention/edema; also raise fracture risk. Edema + weight gain is the trap.

11. Acarbose

WHAT YOU SEE

Brush border enzyme block, postprandial glucose

WHAT IT ENCODES

Alpha-glucosidase inhibitor at brush border — delays carbohydrate digestion, lowers postprandial glucose specifically. GI flatulence common.

BOARD TRAP

Treat acarbose hypoglycemia with oral DEXTROSE/glucose, NOT sucrose — sucrose breakdown is blocked.

12. Front counter — combos

WHAT YOU SEE

FRONT COUNTER, metformin pairings

WHAT IT ENCODES

Metformin is the backbone; combine with SGLT2i or GLP-1 for CV/renal high-risk patients. Add insulin when A1c remains high.

BOARD TRAP

Choosing a DPP-4 add-on in a CKD/HF patient misses guideline-preferred SGLT2i/GLP-1.

13. Hypoglycemia zone

WHAT YOU SEE

HYPOGLYCEMIA ZONE: sulfonylurea, glinide, insulin

WHAT IT ENCODES

Agents that cause hypoglycemia: sulfonylureas, glinides, and insulin (insulin secretagogues + insulin). Counsel on symptoms and rescue glucose.

BOARD TRAP

Metformin, SGLT2i, DPP-4, GLP-1, TZD do not cause hypoglycemia alone — only the secretagogues/insulin do.

14. A1c reduction ranking

WHAT YOU SEE

HbA1c REDUCTION RANKING banner

WHAT IT ENCODES

Metformin gives the most A1c reduction among orals (tied with sulfonylureas). Insulin lowers A1c most overall with no ceiling.

BOARD TRAP

DPP-4 inhibitors have the weakest A1c effect — wrong pick when large reduction is required.
